

Basic Cryptographic Attacks and Types of Cryptography

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1 Basic Cryptographic Attacks

2 Types of Cryptography

Basic Cryptographic Attacks

Definition

A security attack is any malicious action that compromises information or computer systems by exploiting vulnerabilities to gain unauthorized access, steal data, disrupt services, or cause damage, violating core security principles like **confidentiality**, **integrity**, and **availability**.

Remark

Based on the CIA triad we can have the following security breaches-

- Interruption
- Interception
- Modification
- Fabrication

Protocol and Implementation Attack

Man-in-the-Middle (MiTM)

- The attacker transparently relays and manipulates communication while impersonating both parties.
- An attacker secretly intercepts communication between two parties who believe they are directly communicating with each other.
- The attacker can read, modify, or inject messages without being detected.
- Commonly occurs in unauthenticated key exchanges (such as DH Key Exchange Protocol).

Protocol and Implementation Attack

Man-in-the-Middle (MiTM)

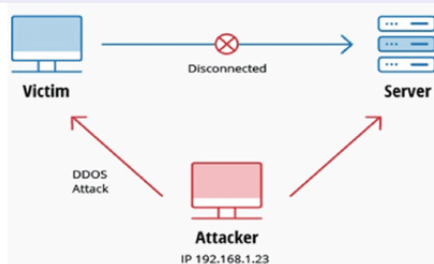
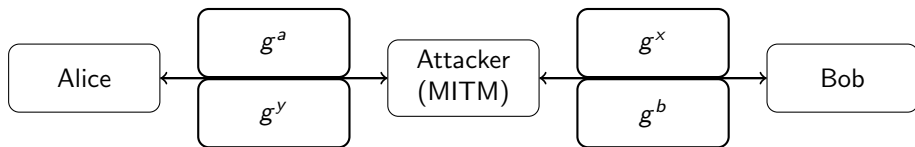


Figure 1: MiTM in Action

Prevention

Use authentication, digital certificates, TLS/HTTPS, and mutual key verification

Diffie–Hellman Man-in-the-Middle Attack



Alice computes $K_1 = (g^y)^a$

Bob computes $K_2 = (g^x)^b$

Key Idea: Attacker establishes two independent keys and relays messages.

Protocol and Implementation Attack

Replay Attack

- An attacker captures a valid message sent by a legitimate user.
- The attacker re-sends (replays) the same message later to trick the system.
- **No decryption** is required; the attack exploits lack of freshness.
- Common in systems without nonces, timestamps, or sequence numbers.

Prevention

Use nonces, timestamps, session IDs, or challenge-response

Protocol and Implementation Attack

Masquerade Attack

- An attacker **impersonates** a legitimate user or system.
- Usually achieved by stolen credentials, keys, or tokens.
- The system cannot distinguish the attacker from the real user.
- Common in systems without nonces, timestamps, or sequence numbers.

Prevention

Strong authentication, multi-factor auth, certificates.

Types of Cryptography

Private Key Cryptography (Symmetric)

- Uses the **same secret key** for both encryption and decryption.
- Faster and computationally efficient.
- Requires a **secure key distribution** mechanism.
- Commonly used for bulk data encryption.
- Examples: AES, DES, Triple DES.

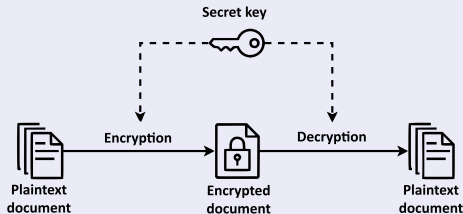


Figure 2: Public Key Cryptography

Types of Cryptography

Public Key Cryptography (Asymmetric)

- Uses a **pair of keys**: public key and private key.
- Public key encrypts data, private key decrypts it.
- Solves the key distribution problem.
- Slower compared to symmetric cryptography.
- Examples: RSA, Diffie–Hellman, ECC.

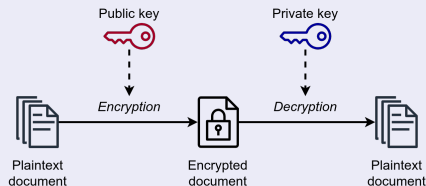


Figure 3: Private Key Cryptography

Symmetric vs Asymmetric Cryptography

- **Key Usage:** Same key vs key pair
- **Speed:** Fast vs Slow
- **Security Focus:** Confidentiality vs Authentication + Key Exchange
- **Typical Use:** Data encryption vs Secure key exchange

Thank You